

QXU5030 Composite Materials

2025B | Prof. Shafique Ahmed

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1. Introduction to Module

■ Assessment

Activity	Score
Coursework(labs)	40%
Final Exam	60%

Week 2N+1: Kahoot! quiz with immediate flash feedback provided after each question.

Week 2N: Quiz delivered via QM+ (or paper-based assignment), with structured written feedback provided within one week.

1.1. Basic Concepts

Anisotropy

A property of a material show different mechanical or physical properties in different directions.

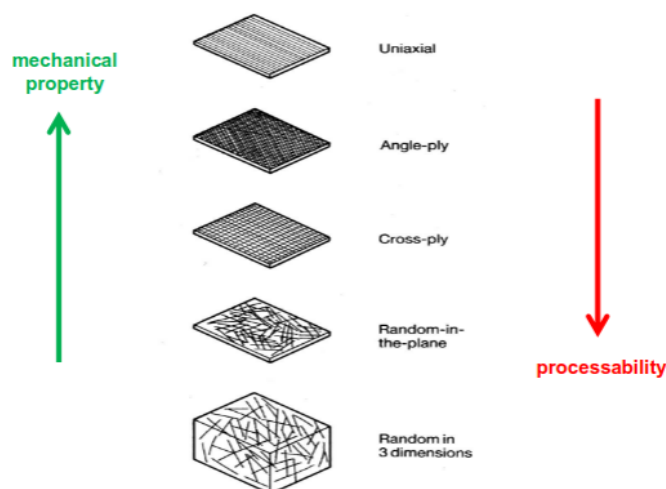
■ Man-Made Composite

Polymer Matrix Composites (PMC's) These are the most common and will be the main area of discussion in this course. Also known as **FRP** - Fibre Reinforced Polymers (or Plastics) - these materials use a polymer-based resin as the matrix, and a variety of fibres such as glass, carbon and aramid as the reinforcement.

Metal Matrix Composites (MMC's) - Increasingly found in the automotive industry, these materials use a metal such as aluminium as the matrix, and reinforce it with fibres such as silicon carbide.

Ceramic Matrix Composites (CMC's) - Used in very high temperature environments, these materials use a ceramic as the matrix and reinforce it with short fibres, or whiskers such as those made from silicon carbide and boron nitride.

■ Type of Composites



1.2. Properties of Composites

■ Factors of Composites Properties

The properties of the composite are determined by:

1. The properties of the fibre
2. The properties of the resin
3. The ratio of fibre to resin (Fibre Volume Fraction)
4. The geometry and orientation of the fibres

■ Specific Strength

Strength-to-weight ratio = tensile strength/density

Indicates how strong a material is relative to its weight.

■ Lightness, or E/ρ^3 and Lightweight Structural Design

E = Young's modulus (stiffness)

E/ρ^3 = key parameter for bending stiffness and buckling resistance in weight-sensitive structures

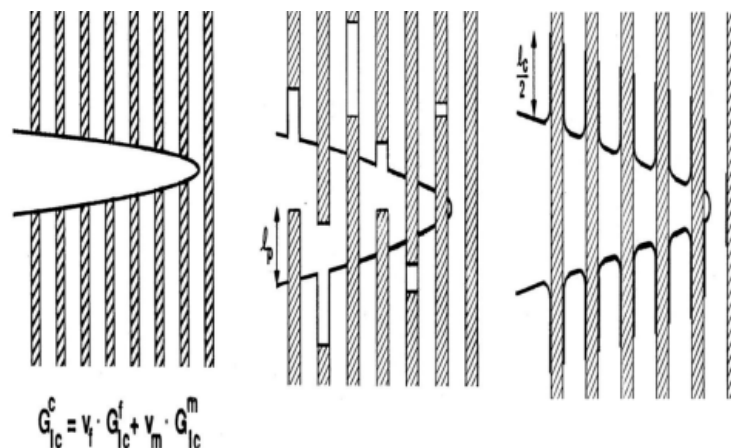
■ Synergism, the "Composite Effect"

Ideally, rule of mixture (RoM) for a physical property P :

$$P_{\text{RoM}} = P_1 V_1 + P_2 V_2 \quad (V_1 + V_2 = 1)$$

However, synergism can make $P_{\text{observed}} \gg P_{\text{RoM}}$

And synergism can modify the mode of fracture by weakening of fibre/matrix interface to prevent brittle fracture.



■ Safety

Composites can absorb more energy per kilo than metals in crash.

2. Fibre Reinforcements

2.1. Introduction

Most fibre have higher mechanical properties than unreinforced resin or matrix.

The four main factors that govern the fibre's contribution are:

1. The basic mechanical properties of the fibre itself.
2. The surface interaction of fibre and resin (the 'interface').
3. The amount of fibre in the composite ('Fibre Volume Fraction').
4. The orientation of the fibres in the composite.

■ Important Synthetic Fibres

Carbon - the elite fibre : stiff(higher than steel), strong, epoxies, good temperature resistance, chemically inert.

Aramid - very strong, but limited to in stiffness relative to carbon.

Glass - strong, similar to aluminium in stiffness, relatively cheap.

2.2. Glass Fibre

■ Glass Fibre Production

Blending quarry products (sand, kaolin, limestone, colemanite) at 1600°C to form liquid glass.

The liquid is passed through micro-fine bushings and simultaneously cooled to produce glass fibre filaments from 8-15 microns in diameter.

The filaments are drawn at high speed (up to 4000 *m/min*) together into a strand or roving.

Then, sizing and form yarns.

■ Glass Fibre Types

E-glass (electrical) - lower alkali content and stronger than A glass (alkali). Good tensile and compressive strength and stiffness, good electrical properties and relatively low cost, but impact resistance relatively poor.

C-glass (chemical) - best resistance to chemical attack. Mainly used in the form of surface tissue in the outer layer of laminates used in chemical and water pipes and tanks.

R or S-glass - higher tensile strength and modulus than E-glass, with better wet strength retention. Developed for aerospace and defence industries, and used in some hard ballistic armour applications.

E-Glass Fibre Forms

Strand. A compactly bundle of filaments, less used.

Yarns. A twisted bundle of filaments, widely used.

Roving. A untwisted, loosely associated bundle.

■ Binders and Sizing

The sizing can:

1. To protect the fibre surface from damage

2. To bind the fibres together in a yarn for ease of processing
3. To impart anti-static properties
4. To provide a chemical link between fibre surface and matrix

2.3. Carbon Fibre

Carbon fibre is produced by the controlled oxidation, carbonisation and graphitisation of carbon-rich organic precursors (most common one is PAN) which are already in fibre form.

Carbon fibres can be grouped as:

1. High strength (HS)
2. Intermediate modulus (IM)
3. High modulus (HM)
4. Ultra high modulus (UHM)

Carbon fibre has the highest specific stiffness of any commercially available fibre, in both tension and compression (in case of HS-fibre) and a high resistance to corrosion, creep and fatigue.

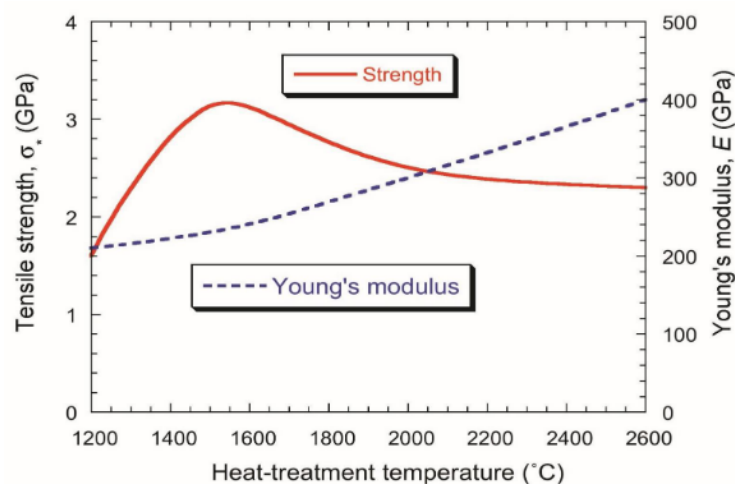
Their impact strength, however, is lower than either glass or aramid, particularly HM and UHM fibres.

■ Carbon Fibre Production by PAN

Tensile modulus of CF can mainly be determined by carbonization/graphitization temperature. Also, the nature of precursors (PAN, pitch) affects on the easiness of reaching a certain level modulus.

Some clothes says "acrylic", is a blended copolymer of polyacrylonitrile, with low purity and short chain.

Here we show the effect of heat treatment on strength and Young's modulus.



■ Carbon Fibre Production by Mesophase Pitch

Heating to

Single BSU $\rightarrow$ Distorted Columns $\rightarrow$ Wrinkled Layers $\rightarrow$ Stiff Layers

Graphitisation between 1500 and 3000 $^{\circ}\text{C}$ gives improved orientation and properties. Heating above 2000 $^{\circ}\text{C}$ gives higher crystallinity and higher (anisotropic) modulus - HM-Carbon.

2.4. Aramid Fibres

Aramid fibre is a man-made organic polymer produced by spinning from a liquid crystalline solution.

1. High strength and low density.
2. Good resistance to impact, abrasion, chemical and thermal degradation.
3. **POOR** compressive strength, and degradate under UV light.